

Pentapalli Charan

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SKILLS

- **Languages:** C++, Python, SQL
- **Frameworks:** Scikit-learn, TensorFlow, PyTorch, Django, React.js, OpenCV
- **Tools/Platforms:** MySQL, PostgreSQL, Git, GitHub, Docker, Jenkins, Jupyter Notebook, Jira, Power BI
- **Core Skills:** DSA, Machine Learning, Computer Vision, NLP
- **Soft Skills:** Critical Thinking, Problem solving, Analytical

INTERNSHIP

Infosys Springboard | Virtual Internship | Certificate

Feb' 26 – Apr' 26

- Developed a full-stack AI-powered image stylization platform (PIXIT) enabling users to upload and transform images using Computer Vision and OpenCV techniques for cartoon effect generation.
- Built REST APIs using FastAPI and integrated PostgreSQL for data management; implemented Docker containerization and automated CI/CD workflows using Jenkins.
- Deployed and orchestrated services using Kubernetes and Terraform while integrating Prometheus and Grafana for monitoring and observability.
- Gained practical exposure to software development, project deployment, testing, DevOps practices.

Tech Stack Used: Python, OpenCV, FastAPI, React.js, PostgreSQL, Docker, Jenkins, Kubernetes, Terraform, Prometheus, Grafana, Nginx

PROJECTS

AI Legal Guidance Assistant | GitHub

Apr' 25

- Built an intelligent legal-assistant chatbot for students using NLP and OpenAI API to interpret queries, and deliver accurate academic and legal guidance.
- Created conversational logic, context tracking, and fallback mechanisms enabling smooth multi-turn interactions, improving clarity, consistency, and user satisfaction.

Tech Stack Used: Python, NLP, Scikit-Learn, OpenAI API

Anomaly Detection in Drone Swarm Technology | GitHub

Nov' 25

- Developed an unsupervised learning system studying natural drone-swarm behaviour and detecting anomalies by analysing formations, movement paths, and communication patterns.
- Implemented clustering algorithms like K-Means and DBSCAN to understand normal swarm dynamics and flag drones that drift, malfunction, or break coordination.
- Designed a practical detection framework improving safety and stability of large-scale swarm operations, valuable for surveillance missions and synchronized drone shows.

Tech Stack Used: Python, Scikit-Learn, K-Means, DBSCAN

TRAINING

Data Structures and Algorithms |Cipher Schools | Certificate

Jun' 25 – Jul' 25

- Strengthened core DSA skills by implementing key data structures such as arrays, linked lists, stacks, queues, trees, graphs, and hash maps using Java.
- Solved algorithmic problems involving recursion, sorting, searching, sliding window, two-pointer methods, and dynamic programming, improving problem-solving accuracy and speed.
- Applied time and space complexity analysis to optimize Java implementations, contributing to cleaner, faster, and interview-ready solutions.

CERTIFICATES

- AI Foundations Associate | Oracle | [Link](#) May'26
- Python for Data Science | NPTEL | [Link](#) Apr' 26
- Introduction to Cloud Computing | NPTEL | [Link](#) Apr' 25
- Introduction to Generative AI | Google Cloud | [Link](#) Aug' 25
- Frontend Web Development | EdYoda | [Link](#) Mar' 24

ACHIEVEMENTS

- **3-Star** Ratings on HackerRank in C++, SQL, Python and Java Mar' 26
- Solved **85+** Coding questions on GeekForGeeks with **135+** score Feb' 26
- Qualified to round-2 in Cyber Security CTF challenge hackathon Feb' 24

EDUCATION

- **Lovely Professional University** Phagwara, Punjab Aug' 23 – Present
- Bachelor of Technology - Computer Science and Engineering; **CGPA: 7.27** Visakhapatnam, Andhra Pradesh Aug' 21 – May' 23
- **Fiitjee Junior College** Palasa, Andhra Pradesh May' 21
- Intermediate; **Percentage: 88.4%**
- **Sri Chaitanya School**
- Matriculation; **Percentage: 99.9%**